



Bapuji Educational Association®
Bapuji Institute of Engineering and Technology, Davanagere
An Autonomous Institute Affiliated to VTU
Department of CSE/ISE and Allied Branches

Course Title	Operating System	Course Code	BCSPCC304
Course Instructor	Prof. Rekha B.H, Dr.Anusha R, Prof. Kaveri C, Prof. Md. Muthair ,Prof. Santhosh P	Semester-Section	III
IA Test No.	I	Max. Marks	30
Date	15.11.2025	Time	3.00 - 4.15 PM

Course Outcome Statements: After the successful completion of the course, the students will be able to	
CO1	Understand the structure and functionality of the operating system.
CO2	Apply appropriate CPU scheduling algorithms for the problems.
CO3	Analyses the various techniques for process synchronization and deadlock handling.
CO4	Apply the various techniques for memory management
CO5	Demonstrate file, secondary storage management strategies and Disk Scheduling Algorithms.

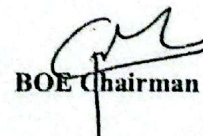
Note: Answer any one full question from each Part.									
Q. No.		Question	Marks	RBT Level	CO				
Part A									
1	A	Define Operating System. Describe interrupt-driven I/O cycle.	6	L2	1				
	B	Explain briefly with an example of how system calls are used.	6	L2	1				
OR									
2	A	List the types of Operating System structures. Explain briefly monolithic and layered OS Kernel.	6	L2	1				
	B	Explain how transition from user to kernel mode work?	6	L2	1				
Part B									
3	A	Outline the process state diagram with all main process states and transitions.	4	L2	2				
	B	Calculate the average waiting time and average turnaround time by drawing the Gantt-chart using FCFS, SJF(non-preemptive), SRTF, RR(q=2ms) algorithms.	8	L3	2				
						Process	Arrival Time	Burst Time	Priority
						P1	0	9	3
						P2	1	4	2
						P3	2	9	1
	P4	3	5	4					
OR									
4	A	Differentiate user-level and kernel-level threads. Compare many-to-one, one-to-one, and many-to-many threading models.	6	L2	2				
	B	Compare and contrast shared memory and message passing models of interprocess communication. Annotate with advantages and disadvantages.	6	L2	2				
Part C									
5	A	Explain Peterson's solution to the critical section problem.	6	L2	3				
	OR								
6	A	Describe the implementation of synchronization hardware using (i) swap instruction (ii) test and set instruction.	6	L2	3				

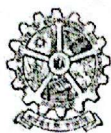
RBT (Revised Bloom's Taxonomy) Levels : Cognitive Domain		
L1 : Remembering	L2 : Understanding	L3 : Applying
L4 : Analysing	L5 : Evaluating	L6 : Creating


Course Instructor /
Coordinator


DAAC
Coordinator


Program Coordinator


BOE Chairman



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Department of CSE/ISE and Allied Branches

Course Title	Operating Systems	Course Code	BCSPCC304
Course Instructors	Prof. Gangamma H, Prof. Kamala R, Prof. Chaitra K J, Prof. Usha C	Semester	III
Test No.	II	Max. Marks	30
Date	09-01-2026	Time	3:45 pm- 5.00pm

Course Outcome Statements: After the successful completion of the course, the students will be able to	
CO1	Understand the structure and functionality of the operating system.
CO2	Apply appropriate CPU scheduling algorithms for the problems.
CO3	Analyse the various techniques for process synchronization and deadlock handling.
CO4	Apply the various techniques for memory management
CO5	Demonstrate file, secondary storage management strategies and Disk Scheduling Algorithms.

Note: Answer any one full question from each Part.							
Q. No.	Question			Marks	RBT Level	CO	
PART A							
1	A	Describe the concept of non-contiguous memory allocation and apply the paging technique with Translation Lookaside Buffer (TLB) to implement it.			6	L3	CO4
	B	Illustrate the working of demand paging with a neat diagram and explain how the operating system handles a page fault during demand paging.			6	L2	CO4
OR							
2	A	Consider the page reference string: 7, 0, 1, 2, 0, 3, 0, 4, 2, 3, 0, 3, 2, 1, 2, 0, 1, 7, 0, 1 Assume a memory with 3-page frames. Determine the number of page faults using the following page replacement algorithms: i) FIFO (First-In First-Out) ii) Optimal iii) LRU (Least Recently Used) Which algorithm is the most efficient?			6	L3	CO4
	B	Describe the different page table structuring techniques used in operating systems with suitable examples.			6	L2	CO4
PART B							
3	A	Suppose that a disk drive has 5000 cylinders, numbered from 0 to 4999. The disk head is currently serving a request at cylinder 143, and the previous request was at cylinder 125. The queue of pending disk I/O requests, in FIFO order, is: 86, 1479, 913, 1774, 948, 1509, 1022, 1750, 130. Starting from the current head position, find the total distance (in cylinders) that the disk head moves to satisfy all the pending requests using the following disk scheduling algorithms: i)FCFS ii)SCAN iii) C-SCAN Assume that the disk head moves initially towards higher numbered cylinders.			6	L3	CO5
	B	List and explain different directory structures used in an operating system with neat diagram			6	L2	CO5
OR							
4	A	Apply the concepts of contiguous, linked, and indexed allocation to explain how files are stored on secondary storage, with neat diagrams.			6	L3	CO5
	B	Outline the general protection model using the Access Matrix concept, explaining the roles of domains and objects.			6	L2	CO5